

ROOTS-AI™ C-05

UI/UX Screen Implementation Specification — Version 1.0.0

Document ID: ROOTS-C05-UIUX-001

Status: APPROVED EXECUTABLE UI/UX RELEASE

Effective date: 21 July 2026

Target: Phase 1 MVP — responsive web application

IMPLEMENTATION CONTROL. This annex fixes the navigation, page inventory, role visibility, layout zones, component behavior, states, responsive rules and UI acceptance evidence for every Phase 1 screen. It is a build specification, not an illustrative mood board. Exact copy remains controlled by C-03 and C-04.

1. Authority and Conflict Rules

Priority	Authority	Controls
1	Master Requirements v3.3	Scope, security, data behavior, delivery and acceptance
2	C-01	Question wording, type, options, required/N/A rules and module order
3	C-02	Scores, calculations, classifications, drivers and deterministic outputs
4	C-03	Report copy contract, 19 sections, report charts, disclaimers and report visual tokens
5	C-04	Website, legal, consent, system and error copy
6	This C-05	Screen composition, navigation, components, states, responsiveness and role visibility

- A screen may not omit a mandatory function because a design library lacks a matching component.
- No implementation may change question order, scoring behavior, legal text or report meaning for visual convenience.
- Where a literal sentence is not repeated here, the exact approved sentence is taken from C-03 or C-04.
- Any material screen, route, modal, permission, navigation item or user-visible state not specified here requires a ROOTS Change Record.

2. Screen Inventory and Navigation Model

ID	Screen	Route	Entry / navigation
PUB-01	Home	/	Public header → logo; all public CTAs
PUB-02	How It Works	/how-it-works	Header Product
PUB-03	Platform	/platform	Header Product
PUB-04	Example Report	/example-report	Header Product
PUB-05	Research	/research	Header Research
PUB-06	Healthcare Professionals	/healthcare-professionals	More menu
PUB-07	Pilot	/pilot	More menu / assessment CTA
PUB-08	About	/about	More menu / footer
PUB-09	Contact	/contact	More menu / footer
PUB-10	Contact Confirmation	/contact?status=sent	Form success
PUB-11	Blog Landing	/blog	More menu / footer
PUB-12	Blog Article Template	/blog/[slug]	Blog card
LEG-01	Privacy	/privacy	Footer Legal
LEG-02	Terms	/terms	Footer Legal
LEG-03	Cookies	/cookies	Footer Legal / consent panel
LEG-04	Medical Disclaimer	/medical-disclaimer	Footer Legal / report
LEG-05	AI Disclaimer	/ai-disclaimer	Footer Legal / report
ASM-01	Assessment Entry	/assessment	Primary CTA
ASM-02	Secure Link Requested	/assessment/check-email	Entry submission
ASM-03	Invalid or Expired Link	/assessment/link-error	Magic-link failure
ASM-04	Consent	/assessment/consent	Valid secure link
ASM-05	Assessment Introduction	/assessment/start	Consent accepted
ASM-06	Module Question Shell	/assessment/[session]/module/[1..13]	Start/resume

ID	Screen	Route	Entry / navigation
ASM-07	Save and Exit	Modal on module shell	Save and Exit
ASM-08	Resume Summary	/assessment/[session]/resume	Valid saved session
ASM-09	Review and Submit	/assessment/[session]/review	Module 13 complete
ASM-10	Report Generating	/assessment/[session]/submitted	Successful submission
ASM-11	Submission Recovery	Same route error state	Generation failure
RPT-01	Report Access	/report/[id]	Secure report link
RPT-02	Interactive Report	/report/[id]	Authorized access
RPT-03	Report Section Navigation	Within /report/[id]	Desktop sidebar / mobile drawer
RPT-04	PDF Preparation	Download action state	Authorized report
RPT-05	Report Access Error	/report/[id] error state	Unauthorized/expired/not found
ADM-01	Admin Sign-in and MFA	/admin	Admin entry
ADM-02	Operations Overview	/admin/dashboard	Admin/Super Admin
ADM-03	Participants	/admin/participants	Admin
ADM-04	Participant Detail	/admin/participants/[id]	Admin
ADM-05	Report Operations	/admin/reports	Admin
ADM-06	Configuration Releases	/admin/configuration	Admin/Super Admin
ADM-07	Configuration Detail	/admin/configuration/[version]	Admin/Super Admin
ADM-08	Audit Viewer	/admin/audit	Super Admin
ADM-09	Research Export	/admin/research-export	Research Admin
ADM-10	Users and Roles	/admin/access	Super Admin
SYS-01	Cookie Consent	Global public overlay	First eligible visit
SYS-02	404	Any unknown route	Router
SYS-03	500 / Service Error	Global boundary	Unhandled service failure
SYS-04	Maintenance	/maintenance or service gate	Planned outage
SYS-05	Offline / Reconnect	Sensitive workflows	Network loss

3. Global Navigation and Shells

Shell	Desktop	Mobile	Mandatory behavior
Public	Logo; How It Works; Platform; Example Report; Research; About; More; Start Assessment	Logo; Start button; hamburger containing every public and legal route	Sticky after 80 px; active route; Skip to content; no participant identity
Assessment	Logo; encrypted-session status; module/progress; Save & Exit	Compact logo; progress; Save icon/menu	No marketing navigation; back/next governed; session-expiry warning
Report	Logo; report metadata; Download PDF; secure sign-out	Logo; section menu; Download; sign-out	Sticky action bar; authorized session only; disclaimer always reachable
Admin	240 px left rail; top bar; environment badge; user menu	Top bar + slide-over navigation	Role-filtered links; MFA state; no hidden-route access by URL

Public header item	Destination / contents
Logo	/
How It Works	/how-it-works

Public header item	Destination / contents
Platform	/platform
Example Report	/example-report
Research	/research
About	/about
More	Healthcare Professionals; Pilot; Blog; Contact
Start Assessment	/assessment

Footer column	Links / content
Product	How It Works; Platform; Example Report; Start Assessment
Research	Research; Pilot; Healthcare Professionals
Company	About; Blog; Contact
Legal	Privacy; Terms; Cookies; Medical Disclaimer; AI Disclaimer
Footer base	ROOTS-AI™ wordmark; copyright; “Educational — Not a Diagnosis”; version/build identifier

4. Roles and Visible UI

Role	Visible screens/actions	Never visible or allowed
Public visitor	Public/legal routes, sample report, contact, assessment entry	Participant report, saved answers, Admin
Participant	Own assessment, own submission status, own interactive/PDF report	Other participants, scoring configuration, audit/export
Admin	Dashboard, participant operations, reports, approved configuration functions	Role changes, audit deletion, raw research identifiers
Research Admin	Research Export only, plus own account controls	Names, emails, full reports, participant operations
Super Admin	Dashboard, configuration, audit, users/roles and approved operations	Audit bypass/deletion, silent published-rule edits
Implementation Team	Only time-bound screens/actions explicitly granted for delivery/support	Production ownership, secondary data use, permanent access

5. Responsive Grid and Visual Tokens

Token	Binding value
Breakpoints	Mobile 360–767 px; Tablet 768–1023 px; Desktop 1024–1439 px; Wide ≥1440 px
Public content width	Maximum 1200 px; reading text maximum 720 px
Grid	Mobile 4 columns / 16 px gutters; Tablet 8 / 24 px; Desktop 12 / 24 px
Spacing	8 px base: 4, 8, 12, 16, 24, 32, 48, 64, 96
Typography	Inter; Arial fallback. H1 48/56 desktop, 34/42 mobile; body 16/26 web; legal body ≥16 px
Primary Navy	#1A2A4A
Soft Navy	#2A4060
Warm White	#FAFAF8
Charcoal	#1A1A1A

Token	Binding value
Muted	#6B7280
Focus	#2563EB with visible 3 px outline
Error	#B42318
Success	#16794B
Cards	12 px radius; 1 px neutral border; shadow only for hierarchy, never as sole affordance
Touch target	Minimum 44×44 px
Motion	150–250 ms; honor prefers-reduced-motion; no essential information in animation

6. Component Contract

Component	Required anatomy	States and rules
Button	Primary, secondary, tertiary/text, destructive	Default, hover, focus, active, disabled, loading; one primary per action region
Text input	Label above; hint below; required marker in text	Default, focus, filled, disabled, error, success; error tied with aria-describedby
Select/radio/checkbox	Visible label and full-row click target	Keyboard selection; never rely on placeholder as label
Question card	Question ID hidden from participant; prompt, help, control, validation	One question at a time on mobile; grouped only where explicitly compatible
Progress	Module n of 13 plus percentage	Text equivalent; no score implication; completed/current/future states
Alert	Info, success, warning, error	Icon + title + text; announced appropriately; persistent for blocking error
Modal	Title, purpose, body, primary/secondary actions	Focus trap; Escape only when safe; return focus to trigger
Toast	Non-blocking save/success confirmation	Auto-dismiss ≥5 s with pause; never sole display of a blocking failure
Card	Heading, body, optional metadata/action	Whole-card link only when one destination; visible focus
Tabs	Maximum 5 visible; otherwise use select/drawer	URL/state preserved; arrow-key navigation
Table	Caption, headers, sorting/filter controls	Horizontal scroll on mobile; stacked cards only if relationships remain clear
Score bar	Label, numeric value, classification, bar	Text and number always present; color not sole meaning
Skeleton	Structure-matched loading placeholder	No false data; replace with retryable error on timeout
Empty state	Reason, safe explanation, permitted next action	No fabricated data; role-aware CTA

7. Public and Legal Screens

PUB-01 — Home

Field	Binding specification
Route / entry	/
Access	Public
Purpose	Explain the product and start assessment
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Header	Public shell
2	Hero	Approved headline, supporting copy, Start Assessment primary CTA, View Example secondary CTA
3	Trust strip	Educational/non-diagnostic language; privacy and deterministic scoring signals
4	Six features	Six report-feature cards from C-04
5	Seven domains	Seven compact domain cards in canonical order
6	How it works	Four-step horizontal/stacked process
7	Example report teaser	Approved report preview with anonymized data
8	Pilot CTA	Eligibility and free beta message
9	Footer	Global public footer

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-02 — How It Works

Field	Binding specification
Route / entry	/how-it-works
Access	Public
Purpose	Explain assessment-to-report workflow
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Intro	Title and boundary statement
2	Four-step process	Assessment; deterministic analysis; governed explanation; report/action
3	What AI does	Language assistance only
4	What AI does not do	No diagnosis or score changes
5	Privacy block	Secure access and data controls
6	CTA	Start Assessment + Example Report

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-03 — Platform

Field	Binding specification
Route / entry	/platform
Access	Public
Purpose	Show Phase 1 and future modules without false availability
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Platform proposition
2	Available now	Assessment and AI Engine cards only
3	Coming soon	Labs, DNA, Microbiome, Wearables, Biological Twin, each visibly labelled Coming Soon
4	Architecture principle	Connected system overview without operational claims
5	CTA	Start Assessment

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-04 — Example Report

Field	Binding specification
Route / entry	/example-report
Access	Public
Purpose	Show approved anonymized 19-section reference
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Intro	Example-only badge and disclaimer
2	Report viewer	All 19 sections using approved sample data
3	Sticky contents	Section links
4	CTA	Start Your Assessment
5	Footer disclaimer	Full educational boundary

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.

Control	Binding behavior
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-05 — Research

Field	Binding specification
Route / entry	/research
Access	Public
Purpose	Present research vision and collaboration channel
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Research vision
2	Principles	Privacy, ethics, consent, transparency
3	Pilot information	Scope and participant protections
4	Collaboration	Approved academic/industry inquiry CTA
5	Boundary	No cohort dashboard or data access

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-06 — Healthcare Professionals

Field	Binding specification
Route / entry	/healthcare-professionals
Access	Public
Purpose	Explain professional relevance without clinician portal claim
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Professional informational positioning
2	Use cases	Conversation support, structured self-report, education
3	Evidence boundary	Not diagnosis, treatment or medical device claim
4	Report overview	What professionals may see only when participant shares it
5	CTA	Contact / Research collaboration

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-07 — Pilot

Field	Binding specification
Route / entry	/pilot
Access	Public
Purpose	Explain beta eligibility and consent-aware join flow
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Free Beta/Pilot proposition
2	Eligibility	Approved inclusion and exclusion statements
3	Benefits	Assessment and educational report
4	Timeline	What happens after joining
5	Privacy/consent	Links to policies and optional research consent
6	FAQs	Approved accordion
7	CTA	Begin Eligibility / Assessment

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-08 — About

Field	Binding specification
Route / entry	/about
Access	Public
Purpose	Explain mission, framework and company
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Mission

Order	Screen zone	Required content and behavior
2	Problem	Medicine Before Symptoms™ framing
3	Framework	Seven domains and governed intelligence
4	Company	ROOTS AI HEALTH SYSTEMS, Inc. information
5	Safety	Educational boundary
6	CTA	Explore Platform / Contact

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-09 — Contact

Field	Binding specification
Route / entry	/contact
Access	Public
Purpose	Route support, research and business inquiries securely
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Intro	Contact purpose
2	Inquiry type	Support, Research, Business radio/select
3	Form	Name, email, subject, message, consent/privacy acknowledgement
4	Safety note	Do not submit emergency or unnecessary health information
5	Submit	Protected, rate-limited action
6	Alternative	Approved business contact if provided

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-10 — Contact Confirmation

Field	Binding specification
Route / entry	/contact?status=sent
Access	Public

Field	Binding specification
Purpose	Confirm safe receipt without exposing message data
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Success	Reference ID, neutral confirmation
2	Next steps	Expected response channel, no guaranteed time unless approved
3	Actions	Return Home; send another inquiry

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-11 — Blog Landing

Field	Binding specification
Route / entry	/blog
Access	Public
Purpose	Display approved articles or professional launch state
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Hero	Blog title and editorial boundary
2	Featured	One featured approved article
3	Grid	Article cards: title, excerpt, date, category
4	Filters	Category only if populated
5	Empty launch	Approved launch-state copy; no broken placeholders

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

PUB-12 — Blog Article Template

Field	Binding specification
Route / entry	/blog/[slug]
Access	Public
Purpose	Render one approved article accessibly
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Breadcrumb	Blog / category / title
2	Article header	Title, subtitle, author, date, reading time
3	Body	Readable 720 px column; semantic headings; citations/links
4	Share	Copy link only unless approved
5	Related	Up to three approved articles
6	Disclaimer	Educational boundary
7	CTA	Assessment or platform, context appropriate

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

LEG-01 — Privacy

Field	Binding specification
Route / entry	/privacy
Access	Public
Purpose	Publish current privacy notice
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Title/meta	Effective date, version, print link
2	Contents	Sticky/inline contents
3	Body	Exact C-04 Privacy copy
4	Rights CTA	Contact/data-rights request channel

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.

Control	Binding behavior
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

LEG-02 — Terms

Field	Binding specification
Route / entry	/terms
Access	Public
Purpose	Publish Terms of Service
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Title/meta	Effective date and version
2	Contents	Anchor navigation
3	Body	Exact C-04 Terms copy
4	Related	Privacy, Medical, AI disclaimer links

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

LEG-03 — Cookies

Field	Binding specification
Route / entry	/cookies
Access	Public
Purpose	Explain cookies and open preferences
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Title/meta	Effective date/version
2	Body	Exact C-04 cookie copy
3	Preferences	Open consent panel
4	Table	Category, purpose, provider, duration where used

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

LEG-04 — Medical Disclaimer

Field	Binding specification
Route / entry	/medical-disclaimer
Access	Public
Purpose	Display complete medical limitation
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Title/meta	Effective date/version
2	Callout	Not diagnosis or medical service
3	Body	Exact C-04 medical disclaimer
4	Emergency	Fixed local-emergency direction
5	Related	Terms and AI Disclaimer

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

LEG-05 — AI Disclaimer

Field	Binding specification
Route / entry	/ai-disclaimer
Access	Public
Purpose	Explain deterministic scoring and governed AI language
Content authority	C-04 controls website/legal copy; C-03 controls example report content.
Responsive rule	Desktop follows 12-column grid; mobile zones stack in listed order; no horizontal body scrolling.

Order	Screen zone	Required content and behavior
1	Title/meta	Effective date/version
2	Summary	AI does not calculate or change scores
3	Body	Exact C-04 AI disclaimer

Order	Screen zone	Required content and behavior
4	Versioning	Current disclaimer/version identifier
5	Related	Medical Disclaimer, Privacy

Control	Binding behavior
Primary actions	Links and CTAs exactly as listed; external links open with clear indication.
Loading / empty / error states	Skeleton only for remote content; approved empty state; inline form errors; global retry for service failure.
Screen acceptance	All zones present; header/footer links work; keyboard order matches visual order; 360/768/1024/1440 px screenshots pass.

8. Assessment Screens and Module Behavior

Module	Participant label	Questions
1	Body Foundations	Q1–Q8
2	Weight & Metabolic History	Q9–Q15
3	Sleep Recovery Index	Q16–Q22
4	Hunger & Satiety Signals	Q23–Q30
5	Stress Load & Inflammation Signals	Q31–Q40
6	Circadian Health Score	Q41–Q45
7	Activity Context	Q46–Q48
8	Biological Safety Signals	Q49–Q51
9	Root Cause Discovery	Q52–Q55
10	Hormonal/Reproductive Context	Q56–Q60
11	Lifestyle & Environment	Q61–Q66
12	Goals & Readiness	Q67–Q71
13	Confidence & Context	Q72–Q73

ASM-01 — Assessment Entry

Field	Binding specification
Route / entry	/assessment
Access	Public
Purpose	Request a secure assessment session
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Header	Minimal brand header
2	Intro	Assessment purpose, estimated time, educational boundary
3	Email	Email field and privacy note

Order	Screen zone	Required content and behavior
4	Eligibility/age	Required launch eligibility acknowledgement if configured
5	CTA	Send Secure Link
6	Links	Privacy, Terms, Medical and AI disclaimers

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-02 — Secure Link Requested

Field	Binding specification
Route / entry	/assessment/check-email
Access	Public after submission
Purpose	Confirm link request without revealing account existence
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Status	Neutral email-sent message
2	Instructions	Check inbox/spam; link expiry
3	Resend	Rate-limited after timer
4	Change email	Return safely

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-03 — Invalid or Expired Link

Field	Binding specification
Route / entry	/assessment/link-error
Access	Public with invalid token
Purpose	Recover without exposing token details
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Error	Expired/invalid generic explanation
2	Action	Request new secure link
3	Support	Contact link

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-04 — Consent

Field	Binding specification
Route / entry	/assessment/consent
Access	Authorized participant
Purpose	Capture mandatory service consent and optional research consent separately
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Identity	Masked email/session status
2	Service consent	Unchecked required checkbox; links to Terms, Privacy, Medical, AI
3	Research consent	Separate unchecked optional checkbox with no service penalty
4	Records	Version/effective date displayed
5	CTA	Agree and Continue; Decline/Exit

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-05 — Assessment Introduction

Field	Binding specification
Route / entry	/assessment/start
Access	Consented participant
Purpose	Prepare participant and explain controls

Field	Binding specification
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Welcome	What the assessment covers
2	Instructions	Answer current experience; use N/A only where offered
3	Privacy	Save/resume and session security
4	Progress preview	13 modules, no scores
5	CTA	Start Module 1

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-06 — Module Question Shell

Field	Binding specification
Route / entry	/assessment/[session]/module/[1..13]
Access	Authorized participant
Purpose	Capture C-01 answers exactly and safely
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Top bar	Module n of 13, percentage, Save & Exit
2	Module intro	Module title and one-sentence purpose
3	Question region	Prompt, help/units, approved control, N/A only when C-01 allows
4	Validation	Inline after blur/Next; summary at region start
5	Navigation	Back and Next; Submit only after Review
6	Autosave	Debounced save status: Saving/Saved/Failed
7	Session warning	Expiry countdown with extend action

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.

Control	Binding behavior
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-07 — Save and Exit

Field	Binding specification
Route / entry	Modal
Access	Authorized participant
Purpose	Confirm save state and safe exit
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Status	Last saved timestamp
2	Body	Explain secure-link resume
3	Actions	Save & Exit primary; Continue Assessment secondary

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-08 — Resume Summary

Field	Binding specification
Route / entry	/assessment/[session]/resume
Access	Authorized participant
Purpose	Show safe progress and resume point
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Welcome back	No health-answer preview on shared screen
2	Progress	Completed modules and current module
3	Save metadata	Last saved time and questionnaire version
4	CTA	Resume Assessment
5	Secondary	Restart requires explicit destructive confirmation

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-09 — Review and Submit

Field	Binding specification
Route / entry	/assessment/[session]/review
Access	Authorized participant
Purpose	Validate completeness and obtain submission confirmation
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Summary	13 modules with Complete/Needs attention
2	Issues	Links to unanswered required questions
3	Optional omissions	Clearly allowed; Q57/Q59 N/A supported
4	Confirmation	Answers become submitted snapshot; educational boundary
5	Actions	Submit Assessment primary; Back to Answers

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-10 — Report Generating

Field	Binding specification
Route / entry	/assessment/[session]/submitted
Access	Authorized participant
Purpose	Confirm submission and show deterministic/report pipeline status
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Success	Submission reference and timestamp
2	Progress	Answers received → scores calculated → report prepared

Order	Screen zone	Required content and behavior
3	Safety	Do not close claim only if true; email/report access expectations
4	Action	Open report when ready; secure sign-out

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

ASM-11 — Submission Recovery

Field	Binding specification
Route / entry	Same route error state
Access	Authorized participant
Purpose	Recover from report-generation failure without duplicate submission
Content authority	C-01 controls every question/control/option; C-04 controls consent and system copy.
Responsive rule	Mobile shows one primary question region at a time; desktop question column max 760 px; progress and navigation remain reachable.

Order	Screen zone	Required content and behavior
1	Status	Answers submitted safely; report not completed
2	Reference	Submission ID and retry state
3	Actions	Retry report preparation; Contact support
4	Boundary	Never ask participant to re-enter answers unless data integrity check fails

Control	Binding behavior
Primary actions	Only the actions listed for the screen; destructive actions require confirmation.
Loading / empty / error states	Autosave, offline, validation, expired session, duplicate-submit and service-failure states use C-04 copy and preserve submitted data.
Screen acceptance	Question IDs/order match C-01; 71 required and 2 optional rules pass; all 13 modules, keyboard flow, save/resume and recovery pass E2E.

9. Report Screens

Order	Section
1	Cover Page
2	Executive Summary
3	ROOTS Biological State™
4	ROOTS Opportunity Score™
5	ROOTS Confidence™
6	Key Drivers

Order	Section
7	Seven-Domain Score Breakdown
8	Biological Triad™
9	Future Projection
10	90-Day Roadmap
11	Nutrition Priorities
12	Action Priorities
13	What Is Going Well
14	Specific Concerns
15	Suggested Laboratory Discussion
16	Participant Answers
17	Biological Card
18	Final Word
19	Medical and AI Disclaimer

RPT-01 — Report Access

Field	Binding specification
Route / entry	/report/[id]
Access	Secure link holder
Purpose	Authorize report access without exposing report existence
Content authority	C-02 controls data; C-03 controls all 19 sections, charts and report copy; C-04 controls access/system copy.
Responsive rule	Desktop report body 800 px plus 240 px contents; mobile single column; score cards stack; charts retain labels and values.

Order	Screen zone	Required content and behavior
1	Verification	Session/token validation
2	Loading	Neutral secure loading
3	Expired	Request new link/contact support
4	Success	Redirect to interactive report

Control	Binding behavior
Primary actions	Section navigation, Download PDF, sign-out and approved contextual links only.
Loading / empty / error states	Null domain, missing content, narrative fallback, PDF failure and access error use approved deterministic/fixed copy.
Screen acceptance	Web/PDF values and versions match canonical JSON; all 19 sections exist; nulls are not imputed; 360/768/1024/1440 visual regression passes.

RPT-02 — Interactive Report

Field	Binding specification
Route / entry	/report/[id]
Access	Authorized participant
Purpose	Render the canonical 19-section report
Content authority	C-02 controls data; C-03 controls all 19 sections, charts and report copy; C-04 controls access/system copy.
Responsive rule	Desktop report body 800 px plus 240 px contents; mobile single column; score cards stack; charts retain labels and values.

Order	Screen zone	Required content and behavior
1	Header	Report ID/version/date; Download PDF; sign-out
2	Cover	Participant display name or Participant; educational badge
3	Summary metrics	Biological State, Opportunity, Recovery Potential, Confidence
4	Section body	All 19 sections in fixed order
5	Footer	Version set, audit reference, disclaimer link

Control	Binding behavior
Primary actions	Section navigation, Download PDF, sign-out and approved contextual links only.
Loading / empty / error states	Null domain, missing content, narrative fallback, PDF failure and access error use approved deterministic/fixed copy.
Screen acceptance	Web/PDF values and versions match canonical JSON; all 19 sections exist; nulls are not imputed; 360/768/1024/1440 visual regression passes.

RPT-03 — Report Section Navigation

Field	Binding specification
Route / entry	Within /report/[id]
Access	Authorized participant
Purpose	Move across long report without changing meaning
Content authority	C-02 controls data; C-03 controls all 19 sections, charts and report copy; C-04 controls access/system copy.
Responsive rule	Desktop report body 800 px plus 240 px contents; mobile single column; score cards stack; charts retain labels and values.

Order	Screen zone	Required content and behavior
1	Desktop	Sticky left contents with 19 numbered links
2	Mobile	Current section button opens full-height section drawer
3	Active state	Intersection observer plus URL hash
4	Return top	Accessible action after section 3

Control	Binding behavior
Primary actions	Section navigation, Download PDF, sign-out and approved contextual links only.

Control	Binding behavior
Loading / empty / error states	Null domain, missing content, narrative fallback, PDF failure and access error use approved deterministic/fixed copy.
Screen acceptance	Web/PDF values and versions match canonical JSON; all 19 sections exist; nulls are not imputed; 360/768/1024/1440 visual regression passes.

RPT-04 — PDF Preparation

Field	Binding specification
Route / entry	Download action state
Access	Authorized participant
Purpose	Generate/download byte-authorized PDF from same canonical report JSON
Content authority	C-02 controls data; C-03 controls all 19 sections, charts and report copy; C-04 controls access/system copy.
Responsive rule	Desktop report body 800 px plus 240 px contents; mobile single column; score cards stack; charts retain labels and values.

Order	Screen zone	Required content and behavior
1	Trigger	Download PDF
2	Preparing	Progress/spinner and cancel UI only if supported
3	Ready	Download starts and confirmation shown
4	Failure	Retry without regenerating scores; support reference

Control	Binding behavior
Primary actions	Section navigation, Download PDF, sign-out and approved contextual links only.
Loading / empty / error states	Null domain, missing content, narrative fallback, PDF failure and access error use approved deterministic/fixed copy.
Screen acceptance	Web/PDF values and versions match canonical JSON; all 19 sections exist; nulls are not imputed; 360/768/1024/1440 visual regression passes.

RPT-05 — Report Access Error

Field	Binding specification
Route / entry	/report/[id] error state
Access	Public/unauthorized
Purpose	Handle not found, expired and forbidden safely
Content authority	C-02 controls data; C-03 controls all 19 sections, charts and report copy; C-04 controls access/system copy.
Responsive rule	Desktop report body 800 px plus 240 px contents; mobile single column; score cards stack; charts retain labels and values.

Order	Screen zone	Required content and behavior
1	Message	Generic secure-access failure
2	Actions	Request new secure link; Contact support; Home

Order	Screen zone	Required content and behavior
3	Security	No participant name, report ID detail, score or existence confirmation

Control	Binding behavior
Primary actions	Section navigation, Download PDF, sign-out and approved contextual links only.
Loading / empty / error states	Null domain, missing content, narrative fallback, PDF failure and access error use approved deterministic/fixed copy.
Screen acceptance	Web/PDF values and versions match canonical JSON; all 19 sections exist; nulls are not imputed; 360/768/1024/1440 visual regression passes.

10. Admin Screens

ADM-01 — Admin Sign-in and MFA

Field	Binding specification
Route / entry	/admin
Access	Privileged users
Purpose	Authenticate privileged users
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Brand	Admin-only identity
2	Credentials/link	Approved provider authentication
3	MFA	Required challenge
4	Recovery	Controlled recovery; no security-question fallback
5	Environment	Visible Staging/Production badge after auth

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-02 — Operations Overview

Field	Binding specification
Route / entry	/admin/dashboard
Access	Admin/Super Admin
Purpose	Show operational state without exposing unnecessary health detail
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.

Field	Binding specification
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Top metrics	Assessments in progress/submitted; reports ready/failed; open incidents
2	Queues	Recent report failures and support actions
3	Release status	Questionnaire/scoring/report template versions
4	System health	Services, backups, last restore test
5	Audit preview	Recent privileged events for Super Admin only

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-03 — Participants

Field	Binding specification
Route / entry	/admin/participants
Access	Admin
Purpose	Find and operate participant records
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Filters	Status, date range, report state; no free-text health-answer search
2	Table	Pseudonymous ID/display name, masked email, status, dates
3	Actions	Open detail; resend approved link; no bulk export
4	Pagination	Server-side; URL-persisted filters

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-04 — Participant Detail

Field	Binding specification
Route / entry	/admin/participants/[id]
Access	Admin
Purpose	Review participant lifecycle and authorized actions
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Identity	Minimum necessary profile/contact
2	Timeline	Invite, consent, save, submit, report events
3	Assessment	Completion and version; raw answers only when role/purpose permits
4	Report	Status, version, secure open/resend
5	Requests	Deletion/access request state
6	Audit	Actions logged; no edit of submitted answers

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-05 — Report Operations

Field	Binding specification
Route / entry	/admin/reports
Access	Admin
Purpose	Monitor generation and recover approved failures
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Filters	Ready/generating/failed; date; version
2	Table	Report ID, participant pseudonym, versions, status, timestamps
3	Detail drawer	Failure code and safe trace reference
4	Actions	Retry governed narrative/PDF only; never edit scores
5	Evidence	Link to audit events

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-06 — Configuration Releases

Field	Binding specification
Route / entry	/admin/configuration
Access	Admin/Super Admin
Purpose	List immutable questionnaire/scoring/template releases
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Version cards/table	Type, version, state, effective date, publisher, checksum
2	Filters	Draft/published/retired; type
3	Actions	View/compare; publish/rollback Super Admin only
4	Warning	Published releases immutable

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-07 — Configuration Detail

Field	Binding specification
Route / entry	/admin/configuration/[version]
Access	Admin/Super Admin
Purpose	Inspect and approve controlled release
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Header	Type/version/state/checksum

Order	Screen zone	Required content and behavior
2	Diff	Current vs candidate field-level comparison
3	Validation	Schema checks and Golden Test result
4	Evidence	Publisher, approval, timestamps
5	Actions	Publish/rollback with reason and MFA re-auth; no silent edit

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-08 — Audit Viewer

Field	Binding specification
Route / entry	/admin/audit
Access	Super Admin
Purpose	Review append-only privileged and security events
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Filters	Actor, action, object, result, date, environment
2	Table	Timestamp, actor, action, object reference, result
3	Detail	Structured metadata with secrets/PHI redacted
4	Export	Approved audit export only
5	Rule	No delete/edit control

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-09 — Research Export

Field	Binding specification
Route / entry	/admin/research-export
Access	Research Admin

Field	Binding specification
Purpose	Create approved de-identified export
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Purpose	Required project/purpose reference
2	Filters	Approved cohort/date/consent criteria only
3	Field preview	Allow-listed pseudonymized fields
4	Risk check	Cell-size/direct-identifier validation
5	Action	Generate expiring encrypted export
6	History	Requester, purpose, checksum, expiry, audit event

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

ADM-10 — Users and Roles

Field	Binding specification
Route / entry	/admin/access
Access	Super Admin
Purpose	Manage named privileged access
Content authority	Master Requirements controls roles, data and audit; C-04 controls system/error copy.
Responsive rule	Desktop optimized; tablet functional; mobile supports urgent review/actions but wide tables may use labeled horizontal scroll.

Order	Screen zone	Required content and behavior
1	Users	Name, email, role, MFA, status, last access
2	Invite	Named account and role
3	Change role	Reason + MFA re-auth + audit
4	Revoke	Immediate access revocation
5	Implementation access	Expiry mandatory; no permanent default

Control	Binding behavior
Primary actions	Only role-authorized actions listed; every write action requires confirmation, reason where stated and audit event.
Loading / empty / error states	Loading skeleton; permission-denied; filtered empty; safe service error; stale-data refresh; session-expiry re-auth.

Control	Binding behavior
Screen acceptance	Positive and negative RBAC tests pass; URL guessing cannot reveal data; every privileged write produces immutable audit evidence.

11. Global System Screens and States

SYS-01 — Cookie Consent

Field	Binding specification
Route / entry	Global public overlay
Access	Eligible public visitors
Purpose	Obtain and manage optional analytics consent
Content authority	C-04 controls exact system and error copy.
Responsive rule	Full viewport; usable at 320 px minimum without clipped primary action.

Order	Screen zone	Required content and behavior
1	Banner	Essential-only default; Accept optional; Reject optional; Preferences
2	Preferences	Categories, purpose, provider; unticked optional default
3	Persistence	Consent version/time
4	Reopen	Footer Cookies link

Control	Binding behavior
Primary actions	Only safe recovery actions listed.
Loading / empty / error states	Every state has visible title, explanation, action and support path where relevant.
Screen acceptance	State can be forced in test environment; keyboard/screen reader behavior and recovery action pass.

SYS-02 — 404

Field	Binding specification
Route / entry	Unknown route
Access	Any
Purpose	Recover from unknown route
Content authority	C-04 controls exact system and error copy.
Responsive rule	Full viewport; usable at 320 px minimum without clipped primary action.

Order	Screen zone	Required content and behavior
1	Message	Page not found
2	Actions	Home; Start Assessment; Contact
3	Security	No route diagnostics

Control	Binding behavior
Primary actions	Only safe recovery actions listed.
Loading / empty / error states	Every state has visible title, explanation, action and support path where relevant.
Screen acceptance	State can be forced in test environment; keyboard/screen reader behavior and recovery action pass.

SYS-03 — 500 / Service Error

Field	Binding specification
Route / entry	Global boundary
Access	Any
Purpose	Recover safely from unexpected failure
Content authority	C-04 controls exact system and error copy.
Responsive rule	Full viewport; usable at 320 px minimum without clipped primary action.

Order	Screen zone	Required content and behavior
1	Message	Unable to complete request
2	Reference	Non-sensitive incident/reference ID
3	Actions	Retry; Home; Contact
4	Logging	Client/server correlation without secrets or health content

Control	Binding behavior
Primary actions	Only safe recovery actions listed.
Loading / empty / error states	Every state has visible title, explanation, action and support path where relevant.
Screen acceptance	State can be forced in test environment; keyboard/screen reader behavior and recovery action pass.

SYS-04 — Maintenance

Field	Binding specification
Route / entry	Maintenance gate
Access	Any
Purpose	Communicate planned unavailability
Content authority	C-04 controls exact system and error copy.
Responsive rule	Full viewport; usable at 320 px minimum without clipped primary action.

Order	Screen zone	Required content and behavior
1	Status	Maintenance message
2	Timing	Expected return only when approved
3	Actions	Refresh; public status/contact if configured
4	Security	Admin bypass never public

Control	Binding behavior
Primary actions	Only safe recovery actions listed.
Loading / empty / error states	Every state has visible title, explanation, action and support path where relevant.
Screen acceptance	State can be forced in test environment; keyboard/screen reader behavior and recovery action pass.

SYS-05 — Offline / Reconnect

Field	Binding specification
Route / entry	Assessment/report/admin overlays
Access	Authorized user
Purpose	Protect work during network loss
Content authority	C-04 controls exact system and error copy.
Responsive rule	Full viewport; usable at 320 px minimum without clipped primary action.

Order	Screen zone	Required content and behavior
1	Banner	Offline status
2	Assessment	Keep unsent current input locally in memory only; retry save; warn before exit
3	Submitted/report	Never fabricate success; retry status
4	Admin	Disable writes until fresh authorization/state
5	Recovery	Reconcile and confirm saved timestamp

Control	Binding behavior
Primary actions	Only safe recovery actions listed.
Loading / empty / error states	Every state has visible title, explanation, action and support path where relevant.
Screen acceptance	State can be forced in test environment; keyboard/screen reader behavior and recovery action pass.

12. Interaction, Validation and Persistence Rules

Area	Binding rule
Navigation	Browser Back/Forward preserves public state and safe filters; assessment Back uses saved session state; no accidental resubmission.
Autosave	Save after stable answer change and before Next; show Saving/Saved/Failed; never show Saved without server acknowledgement.
Validation	Validate against C-01; show inline error and top summary on submit; focus first invalid control; preserve all valid values.
Submission	Idempotency key prevents duplicate answer/report records; disable repeated submit while pending; recovery checks server status.
Sessions	Warn before expiry; successful extension preserves state; expired sessions clear sensitive view and require secure re-entry.
Modals	Focus trap, labelled dialog, safe Escape behavior and focus return.

Area	Binding rule
Destructive actions	Explain effect; require explicit confirmation; typed confirmation only for irreversible high-risk actions.
URLs	No Personal Data, answers, score values or secrets in query strings, hashes or analytics events.
Analytics	Disabled on assessment, report and Admin; public optional analytics load only after applicable consent.
Printing	Legal pages printable; report printing routes to approved PDF; Admin tables do not expose hidden columns.

13. Accessibility and Content Quality

- Meet WCAG 2.2 AA for all public, participant and Admin screens.
- One H1 per screen; semantic heading order; landmark regions; Skip to content on every shell.
- All functions operable by keyboard; no keyboard trap; logical focus order; visible focus on every interactive element.
- Form labels, errors and help text are programmatically associated; required state is communicated in text and code.
- Charts include label, numeric value, classification and text explanation; color is never the sole carrier of meaning.
- Contrast: normal text $\geq 4.5:1$, large text $\geq 3:1$, non-text UI/focus $\geq 3:1$.
- Zoom to 200% and reflow at 320 CSS px without loss of function or two-dimensional scroll except approved data tables.
- Announcements use restrained live regions; saving and loading do not cause focus jumps.
- Plain-language copy and exact legal/medical wording must not be shortened by responsive design.

14. Screen Evidence and Acceptance

ID	Acceptance criterion
UI-001	Every screen ID in Section 2 has a working Staging route/state and test evidence.
UI-002	Public header, mobile menu and footer reach every listed route with correct active state.
UI-003	Role-based navigation and direct URL access pass positive and negative tests.
UI-004	All 73 questions render from C-01 in exact order across 13 modules.
UI-005	Save/resume, expiry, offline and failed-save recovery preserve confirmed answers.
UI-006	Review/submit blocks missing required answers and accepts the two optional records.
UI-007	Duplicate submission is prevented and report-generation recovery does not duplicate scores.
UI-008	Interactive report contains exactly 19 sections and matches PDF/canonical JSON.
UI-009	Null/missing states use approved language and never impute a value.
UI-010	Admin tables, actions and data fields match role permissions and audit requirements.
UI-011	Published configuration cannot be silently edited; publish/rollback produces evidence.
UI-012	Analytics requests are absent on assessment, report and Admin routes.
UI-013	360, 768, 1024 and 1440 px reference screenshots are delivered for every screen/state family.
UI-014	Keyboard-only, screen-reader smoke, contrast, zoom/reflow and automated accessibility reports pass.
UI-015	Chrome, Edge, Firefox, Safari and current mobile Safari/Chrome compatibility passes.
UI-016	No placeholder, lorem ipsum, broken route, unlabeled Coming Soon control or unapproved copy remains.
UI-017	Loading, empty, error, success, unauthorized and offline states are force-tested where applicable.
UI-018	Performance targets in the Master Requirements pass on agreed test conditions.

ID	Acceptance criterion
UI-019	ROOTS approves the screen inventory, responsive evidence and visual-regression baseline during UAT.

15. Required UI/UX Handover Artifacts

#	Artifact
1	Implemented responsive component library and documented tokens
2	Storybook or equivalent component/state catalog accessible in ROOTS-controlled environment
3	Route-to-screen-ID inventory and role matrix
4	Reference screenshots at 360, 768, 1024 and 1440 px for every screen/state family
5	Visual-regression test baseline and results
6	Accessibility audit, keyboard test and screen-reader smoke evidence
7	Browser/device compatibility report
8	All icons, illustrations, fonts and licenses used
9	Editable source assets and export settings
10	Content mapping showing C-03/C-04 source for every user-visible block
11	Known Medium/Low visual defects and approved disposition

No separate Figma file is required when the implemented component catalog, screen evidence and editable source assets fully satisfy this annex. If the Implementation Partner uses Figma, the editable file must be transferred to a ROOTS-owned workspace and may not replace or weaken this binding specification.